

Hazard Knowledge Compiler

Literature Corpus D

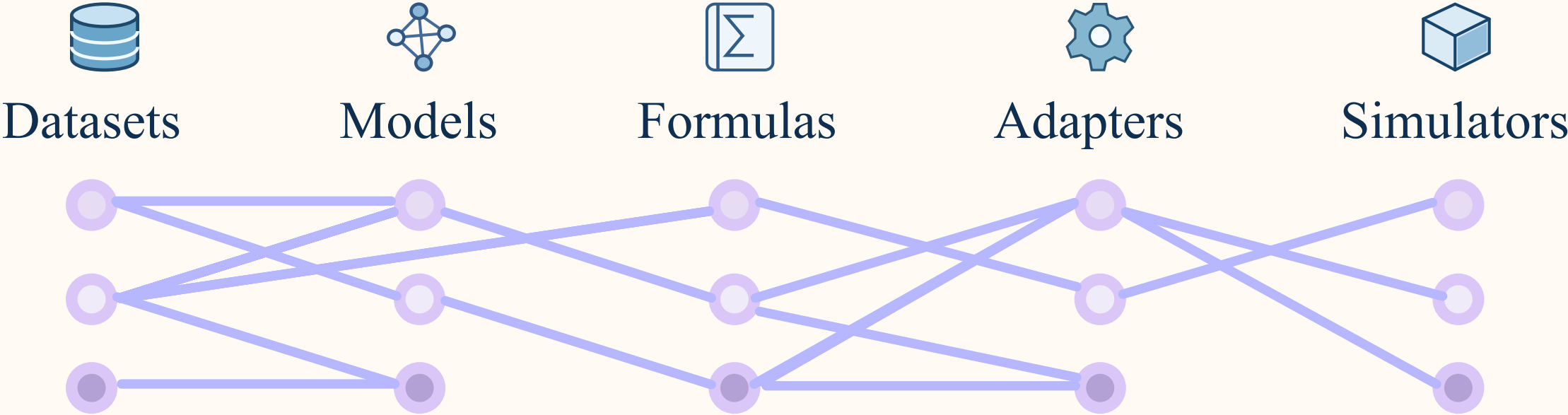
- papers
- reports
- guidelines

Extract
Ground
Canonicalize

Scientific Conditions K

- mechanisms
- assumptions
- evidence provenance

Hazard Capability Graph



User Task q

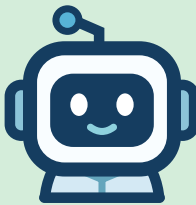
The Thomas Fire burned the hills above Montecito. If a big storm hits, how bad could the mudslides get?

Available State (sk)

- available data
- available capabilities

Candidate Routes rt

- empirical
- physical
- hybrid models
- earth observation
- machine learning



Hazard Weaver Agent

Core Tools

- Observe task & state
- Query HKC / HCG
- Reason eligible routes
- Select reasonable route
- Execute capabilities
- Update state

Route Eligibility

Scientific Applicability
 $Asci(r,q;K)$

Capability Reachability
 $Acap(r,sk;G)$

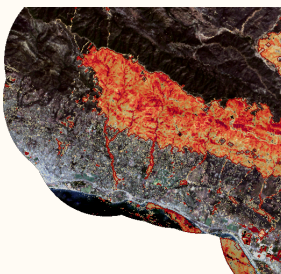
Route Execution

commit route rk
instantiate workflow π
execute capabilities ak

scientific artifact y



Inputs:
Available satellite view
after the Thomas Fire



HWA: How badly it burn?
Route1: Prithvi
yellow = burn severity



HWA: Where debris flow?
Route2: Gorr WEST
purple = sediment source
cyan = likely flow path

Hazard
Weaver
Benchmark

Hazard $\times$ Task Atlas

- flood
- wildfire
- drought
- landslide

- heatwave
- earthquake
- tropical cyclone

post-fire debris flow
coastal compound flood
earthquake-landslide
multi-peril impact

Forecast
Assess
Simulate
Attribute

Route-aware Evaluator

- execution trajectory τ
- scientific artifact y
- Artifact Correctness
- Executed Route Validity
- Recovery / Abstention